(c) (i) • Stand the bottle upright on the bench.

• The height of the water in the bottle is h, as shown in Fig. 2.5.

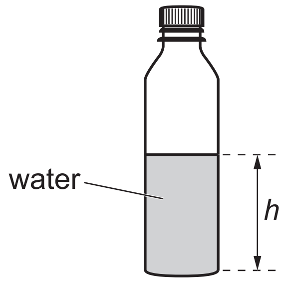

Fig. 2.5

Measure and record h.

h = ...

• The time for the bottle to roll distance L on the rules is t.

Take measurements to determine t.

t = ...
[3]

(ii) A value for the acceleration a of the bottle is given by

$$a = \frac{2L}{t^2}.$$

Calculate a.

a = ... [1]

(iii) Justify the number of significant figures that you have given for your value of a.

....

....

.... [1]

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